

UCS503P .

College Mentorship Management System

A Digital Platform for Structured Faculty–Student Mentorship

Department of Computer Science and Engineering • Thapar Institute of Engineering and Technology

JS

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Frontend Lead & UI/UX

Responsive React SPA, 3 role dashboards, design system & accessibility

AB

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Backend Lead & API Architecture

Node.js & Express REST APIs, JWT auth, matching engine, conflict-free booking

HS

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Database Lead & Integration

MongoDB schemas, data stores, meeting-notes logging & pilot validation

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Introduction

Why a purpose-built mentorship platform is needed — and what we built instead

01

Problem

- Mentorship requests get buried under ad-hoc student emails, with no central record.
- Office hours are often unscheduled, forgotten, or cancelled without notice.
- Meeting outcomes are never logged — at-risk students are flagged too late.

02

Our Solution

- A unified three-tier web app replacing ad-hoc email coordination.
- Auto-matches students to mentors by department and focus area.
- Conflict-free scheduling, persistent meeting logs, and admin analytics.

Technologies Used

Layer	Technology
Frontend	React.js + Vite
Backend	Node.js + Express.js
Database	MongoDB Atlas + Mongoose
API	REST API + JWT Auth
Matching Logic	Rule-based allocation engine
Deployment	Render / Vercel (free tier)

System Objectives & Features

01

Intelligent Allocation

- Department & focus-area alignment
- First-available, transparent matching
- Admin override for manual rebalancing

02

Conflict-Free Scheduling

- Faculty publish office-hour slots
- Students book in one click
- Server-side conflict prevention

03

Meeting Logging & Notes

- Post-meeting summary notes
- Timestamped student audit trail
- Access restricted to mentor & admin

04

Role-Tailored Dashboards

- Student: mentor, bookings, log
- Faculty: roster, schedule, notes
- Admin: metrics & reassignments

Prototype in Action

Live screenshots from the working Student Workspace — React frontend connected to the Node/Express API

Student Dashboard

At-a-glance mentor status, meeting count, and quick actions

The screenshot shows the 'Student Dashboard' of the 'Mentorship Management System'. The interface includes a dark sidebar with navigation links: 'Dashboard', 'My Mentor', 'Request Mentor', 'Book Meeting', and 'Meeting History'. The main content area features a header with 'UNIVERSITY PORTAL' and 'Mentorship Management System', and a user profile 'jessica Student - Csed' with a 'Logout' button. Below the header, the dashboard displays 'Welcome, jessica csed' and three key metrics: 'Assigned Mentor' (None), 'Meetings' (1), and 'Semester' (5). At the bottom, there are four action buttons: 'View Mentor' (See assigned mentor profile and availability), 'Request Mentor' (Submit a mentor allocation request), 'Book Meeting' (Schedule time with a mentor), and 'View History' (Review agendas, notes, and statuses).

Book Meeting

Conflict-free slot booking with mentor, date/time & agenda

The screenshot shows the 'Book Meeting' form in the 'Mentorship Management System'. The interface includes a dark sidebar with navigation links: 'Dashboard', 'My Mentor', 'Request Mentor', 'Book Meeting', and 'Meeting History'. The main content area features a header with 'UNIVERSITY PORTAL' and 'Mentorship Management System', and a user profile 'jessica Student - Csed' with a 'Logout' button. Below the header, the form is titled 'STUDENT Book Meeting' and contains three input fields: 'Mentor' (a dropdown menu showing 'Dr. Amit Kumar - CSE'), 'Date and Time' (a date and time picker showing 'dd/mm/yyyy, --:-- --'), and 'Agenda' (a text area). A blue 'Book Meeting' button is located at the bottom of the form.

Prototype in Action

Meeting History

The screenshot displays the 'University Portal Mentorship Management System' (MMS) Student Workspace. The left sidebar contains navigation links: Dashboard, My Mentor, Request Mentor, Book Meeting, and Meeting History (highlighted). The main content area shows the 'STUDENT Meeting History' section. It features a table with columns: DATE, MENTOR, AGENDA, STATUS, and NOTES. A single meeting entry is listed with the date '18 Sept 2026, 02:14', mentor 'Dr. Amit Kumar', agenda 'discuss regarding a research paper', status 'Scheduled' (indicated by a blue pill), and notes 'No notes'. The top right of the page shows the user 'jessica' (Student · Csed) and a 'Logout' button.

DATE	MENTOR	AGENDA	STATUS	NOTES
18 Sept 2026, 02:14	Dr. Amit Kumar	discuss regarding a research paper	Scheduled	No notes

1

Post-meeting summary notes

2

Timestamped student audit trail

3

Access restricted to mentor & admin

4

Status tracked — scheduled, completed, or cancelled

Methodology

Request-to-booking pipeline behind every mentorship allocation

1

Mentor Request Submitted

2

Request Validation

3

Conflict Detection (Schedule)

4

Allocation / Resolution Logic

5

Booking Confirmed & Logged

Constraints Considered

- Department & focus-area match
- Faculty availability & office-hour slots
- Existing bookings (conflict window)
- Current mentor workload / capacity

Allocation Approach

First-available rule-based matching, ranked by department fit and current mentor load — with a manual admin override for edge cases.

MVP note: the pilot database seeds a fixed set of CSE faculty mentors and department mappings.

MongoDB — Users & Allocation Store

Week Implementation Roadmap-12

An agile, modular delivery plan ensuring on-time completion and pilot validation

WEEKS 1-2

Requirements & Schema Design

Architecture specs, ER diagrams, repo setup

WEEKS 3-5

Authentication & Allocation

JWT secure auth, student-mentor matching engine

WEEKS 6-8

Scheduling & Meeting Notes

Office-hour calendar, booking conflict engine, logs

WEEKS 9-10

Dashboards & Integration

Complete React UI for Student, Faculty & Admin

WEEK 11

Pilot Testing

Controlled pilot in CSE Dept, feedback collection

WEEK 12

Buffer & Final Report

Feedback incorporation, documentation, demo

Future Enhancements

1

Automated Notifications

Email/push alerts for new bookings, cancellations, and mentor reassignments.

2

Calendar Integration

Two-way sync with Google Calendar / Outlook for office-hour slots.

3

AI-Assisted Matching

Weight mentor–student pairing by interests and past feedback, not just availability.

4

Admin Analytics Export

Downloadable engagement reports and CSV/PDF exports for department heads.

5

Mobile App & Wider Rollout

Native companion app and deployment across all departments campus-wide.